

Notes on Schwinger-Keldysh Formalism in Condensed Matter Systems

2026.1.9 testing

Part 1: Review of the Closed Time Contour

Mainly based on Hong Liu's lecture (TASI 2017), 1805.09331

1.1 Time Evolution

In the Schrödinger picture, the equation of motion is:

$$i \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle. \quad (1)$$

The definition of the time evolution operator is given by:

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle. \quad (2)$$

Substituting (2) into (1), we obtain:

$$i \frac{\partial}{\partial t} U(t, t_0) = H(t) U(t, t_0). \quad (3)$$

If the Hamiltonian is time-independent, $H(t) = H_0$, the solution to (3) is simply:

$$U(t, t_0) = e^{-iH_0(t-t_0)}. \quad (4)$$

More generally, using the Baker-Campbell-Hausdorff (BCH) formula, the solution is:

$$U(t, t_0) = \mathcal{T} \left(e^{-i \int_{t_0}^t H(\tau) d\tau} \right), \quad t > t_0, \quad (5)$$

where $\mathcal{T}$ denotes time-ordering (operators at later times are placed to the left). For the proof, see **Appendix A**.

For **In-Out** processes (scattering problems), such as the S-matrix:

$$\langle \text{out}, t_f | \text{in}, t_i \rangle = \langle 0 | U(t_f, t_i) | 0 \rangle. \quad (6)$$

For **In-In** processes, such as the expectation value in a pure state:

$$\langle O(t) \rangle \equiv \langle \psi(t) | O(t) | \psi(t) \rangle = \langle \psi(t_0) | U^\dagger(t, t_0) O(t) U(t, t_0) | \psi(t_0) \rangle. \quad (7)$$

The Hermitian conjugate of $U(t, t_0)$ is:

$$U^\dagger(t, t_0) = \bar{\mathcal{T}} \left(e^{i \int_{t_0}^t H(\tau) d\tau} \right) = \bar{\mathcal{T}} \left(e^{-i \int_t^{t_0} H(\tau) d\tau} \right) = U^{-1}(t, t_0) = U(t_0, t), \quad (8)$$

where $\bar{\mathcal{T}}$ denotes anti-time ordering.

We define the contour γ by setting $t_0 = t_i = -\infty$ and $t < t_f = +\infty$. We also define the forward branch $\gamma_+ : t_i \rightarrow t_f$ and the backward branch $\gamma_- : t_f \rightarrow t_i$. There are two evolution cases required to evaluate $\langle O(t) \rangle$:

1. $\gamma : t_i \rightarrow t \equiv t_+ \rightarrow t_f \rightarrow t_i$

$$\langle O(t) \rangle = \langle \psi(t_i) | U^\dagger(t_f, t_i) U(t_f, t_+) O(t_+) U(t_+, t_i) | \psi(t_i) \rangle. \quad (9)$$

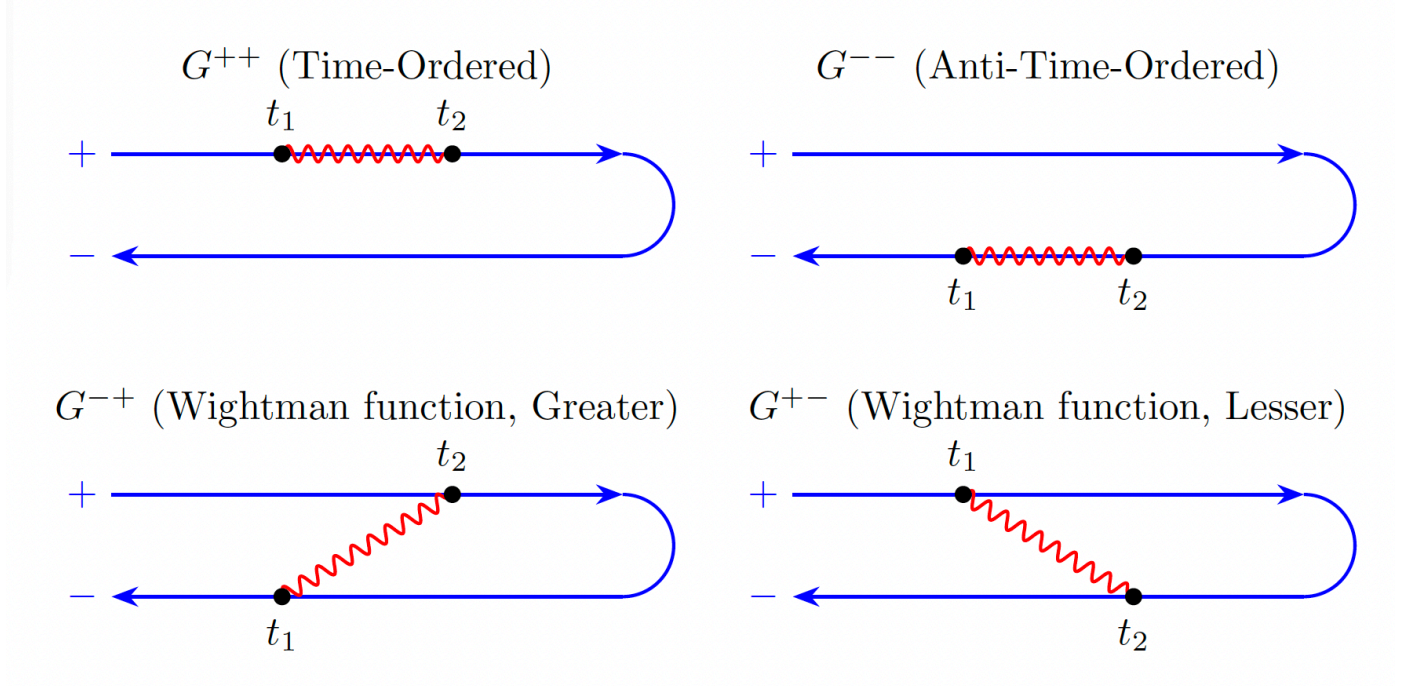
$$2. \quad \gamma : t_i \rightarrow t_f \rightarrow t \equiv t_- \rightarrow t_i$$

$$\langle O(t) \rangle = \langle \psi(t_i) | U^\dagger(t_-, t_i) O(t_-) U^\dagger(t_f, t_-) U(t_f, t_i) | \psi(t_i) \rangle. \quad (10)$$

These can be combined using the contour-ordering operator $\mathcal{P}$:

$$\langle O(t) \rangle = \langle \psi(t_i) | \mathcal{P} \left(e^{-i \int_\gamma H(\tau) d\tau} O(t_\pm) \right) | \psi(t_i) \rangle. \quad (11)$$

We must double the operators to distinguish their location on the path. To illustrate this argument, we take an example, i.e. there are four different two-point Green's functions for O :



$$G_{ab} = \langle \mathcal{P}(O_a(t_1)O_b(t_2)) \rangle = \begin{cases} \langle \mathcal{T} O_+(t_1) O_+(t_2) \rangle, & a = +, b = + \\ \langle \bar{\mathcal{T}} O_-(t_1) O_-(t_2) \rangle, & a = -, b = - \\ \langle O_-(t_1) O_+(t_2) \rangle, & a = -, b = + \\ \langle O_-(t_2) O_+(t_1) \rangle, & a = +, b = - \end{cases} \quad (12)$$

In the interaction picture, we split the Hamiltonian:

$$H(t) = H_0 + \delta H(t), \quad \delta H(t) \sim -O(t)J(t). \quad (13)$$

We define the interaction picture state as:

$$|\psi(t)\rangle_I \equiv U_0^\dagger |\psi(t)\rangle_S, \quad U_0 = e^{-iH_0\Delta t}, \quad (14)$$

and the operator as:

$$\hat{O}_I(t) = U_0^\dagger O_S(t) U_0 = U_I^\dagger(t, t_0) O_I(t_0) U_I(t, t_0). \quad (15)$$

The time evolution for the state is:

$$\begin{aligned}
i \frac{d}{dt} |\psi(t)\rangle_I &= i \frac{d}{dt} (U_0^\dagger |\psi(t)\rangle_S), \\
&= i \frac{dU_0^\dagger}{dt} |\psi(t)\rangle_S + i U_0^\dagger \frac{d|\psi(t)\rangle_S}{dt}, \\
&= -H_0 U_0^\dagger |\psi(t)\rangle_S + U_0^\dagger (H_0 + \delta H(t)) |\psi(t)\rangle_S, \\
&= \delta H(t) |\psi(t)\rangle_I.
\end{aligned} \tag{16}$$

Thus (omitting the subscript I for the state):

$$|\psi(t)\rangle = U_I(t, t_0) |\psi(t_0)\rangle, \quad U_I(t, t_0) = \mathcal{T} \left(e^{-i \int_{t_0}^t \delta H(\tau) d\tau} \right). \tag{17}$$

The evolution for the operator is:

$$\begin{aligned}
\frac{d}{dt} O_I(t) &= i H_0 e^{i H_0 t} O_S(t) e^{-i H_0 t} - e^{i H_0 t} O_S(t) H_0 e^{-i H_0 t} + e^{i H_0 t} \frac{\partial}{\partial t} O_S(t) e^{-i H_0 t}, \\
&= -i [O_I(t), H_0] + \left(\frac{\partial O_S(t)}{\partial t} \right)_I.
\end{aligned} \tag{18}$$

Equation (11) can then be expanded as:

$$\begin{aligned}
\langle O(t) \rangle &= \langle \psi(t_i) | \mathcal{P} \left(e^{-i \int_{\gamma} \delta H(\tau) d\tau} O(t_{\pm}) \right) | \psi(t_i) \rangle \\
&= \langle \psi(t_i) | \mathcal{P} \left(e^{-i (\int_{\gamma_+} \delta H(\tau) d\tau + \int_{\gamma_-} \delta H(\tau) d\tau)} O(t_{\pm}) \right) | \psi(t_i) \rangle \\
&= \langle \psi(t_i) | \mathcal{P} \left(e^{-i \int_{t_i}^{t_f} \delta H(\tau_+) d\tau_+ - i \int_{t_f}^{t_i} \delta H(\tau_-) d\tau_-} O(t_{\pm}) \right) | \psi(t_i) \rangle \\
&= \langle \psi(t_i) | \mathcal{P} \left(e^{i \int_{t_i}^{t_f} O_+(\tau_+) J_+(\tau_+) d\tau_+ + i \int_{t_f}^{t_i} O_-(\tau_-) J_-(\tau_-) d\tau_-} O(t_{\pm}) \right) | \psi(t_i) \rangle \\
&= \langle \psi(t_i) | \mathcal{P} \left(e^{i \int_{t_i}^{t_f} (O_+(\tau) J_+(\tau) - O_-(\tau) J_-(\tau)) d\tau} O(t_{\pm}) \right) | \psi(t_i) \rangle.
\end{aligned} \tag{19}$$

For a mixed state described by a density matrix:

$$\langle O(t) \rangle_\rho = \sum_j \omega_j \langle \psi_j(t) | O(t) | \psi_j(t) \rangle, \quad \text{with } \sum_j \omega_j = 1. \tag{20}$$

We define the density operator:

$$\rho(t) = \sum_j \omega_j |\psi_j(t)\rangle \langle \psi_j(t)|, \quad \text{with } \text{Tr}(\rho) = 1. \tag{21}$$

Its time evolution is given by:

$$\rho(t) = U_I(t, t_0; O_+) \rho(t_0) U_I^\dagger(t, t_0; O_-). \tag{22}$$

Then (20) becomes:

$$\begin{aligned}
\langle O(t) \rangle_\rho &= \text{Tr}(\rho(t)O(t)), \\
&= \text{Tr}\left(U_I(t, t_0; O_+)\rho(t_0)U_I^\dagger(t, t_0; O_-)O(t)\right), \\
&= \text{Tr}\left(\rho(t_0)U_I^\dagger(t, t_0; O_-)O(t)U_I(t, t_0; O_+)\right), \\
&= \text{Tr}\left(\rho(t_0)\bar{\mathcal{T}}\left(e^{-i\int_{t_0}^t O_-(\tau)J_-(\tau)d\tau}\right)O(t)\mathcal{T}\left(e^{i\int_{t_0}^t O_+(\tau)J_+(\tau)d\tau}\right)\right), \\
&= \dots \\
&= \text{Tr}\left(\rho_0\mathcal{P}\left(e^{i\int_{t_i}^{t_f}(O_+(\tau)J_+(\tau)-O_-(\tau)J_-(\tau))d\tau}O(t_\pm)\right)\right), \quad \rho(t_i) = \rho(t_0) \equiv \rho_0.
\end{aligned} \tag{23}$$

We define the **connected generating functional** W :

$$e^{W[J_+, J_-]} \equiv \text{Tr}\left(\rho_0\mathcal{P}\left(e^{i\int_{t_i}^{t_f}(O_+(\tau)J_+(\tau)-O_-(\tau)J_-(\tau))d\tau}\right)\right). \tag{24}$$

In terms of path integrals (See **Appendix B** for a review on path integrals):

$$e^{W[J_+, J_-]} = \int D\xi_a D\xi_b D\xi_c \langle \xi_b | \rho_0 | \xi_a \rangle \int_{\varphi_+(t_i)=\xi_a}^{\varphi_+(t_f)=\xi_c} D\varphi_+ \int_{\varphi_-(t_i)=\xi_b}^{\varphi_-(t_f)=\xi_c} D\varphi_- e^{iS[\varphi_+, J_+] - iS[\varphi_-, J_-]}. \tag{25}$$

1.2 Effective Field Theory

After integrating out the UV fields in (25), we write the effective action as $(J_+(t) \rightarrow \int d\vec{x} J_1(t, \vec{x})$ and $J_-(t) \rightarrow \int d\vec{x} J_2(t, \vec{x}))$:

$$e^{W[J_1, J_2]} = \int_\epsilon D\chi_1 D\chi_2 e^{iI[\chi_1, J_1; \chi_2, J_2]}. \tag{26}$$

The effective action I satisfies three key properties:

1. $I^*[\chi_1, J_1; \chi_2, J_2] = -I[\chi_2, J_2; \chi_1, J_1]$.
2. $I[\chi_1 = \chi_2] = 0$.
3. $\text{Im}[I] \geq 0$.

Proof:

From the definition (24):

$$e^{W[J_1, J_2]} = \text{Tr}(\rho_0 U^\dagger(t_f, t_i; O_2)U(t_f, t_i; O_1)) = \text{Tr}(\rho_0 U^\dagger(+\infty, -\infty; O_2)U(+\infty, -\infty; O_1)). \tag{27}$$

Taking the complex conjugate:

$$\begin{aligned}
e^{W^*[J_1, J_2]} &= \text{Tr}(U^\dagger(+\infty, -\infty; O_1)U(+\infty, -\infty; O_2)\rho(t_0)), \\
&= \text{Tr}(\rho_0 U^\dagger(+\infty, -\infty; O_1)U(+\infty, -\infty; O_2)), \\
&= e^{W[J_2, J_1]}.
\end{aligned} \tag{28}$$

This implies $I^*[\chi_1, J_1; \chi_2, J_2] = -I[\chi_2, J_2; \chi_1, J_1]$. Furthermore, $e^{W[J_1=J_2]} = \text{Tr}(\rho_0) = 1$, which implies $I[\chi_1 = \chi_2] = 0$.

Using the Cauchy-Schwarz inequality $|\langle a|b\rangle|^2 \leq \langle a|a\rangle\langle b|b\rangle$:

$$\begin{aligned}
|e^{W[J_1, J_2]}| &= \left| \sum_j \omega_j \langle \psi_j | U^\dagger(+\infty, -\infty; O_2) U(+\infty, -\infty; O_1) | \psi_j \rangle \right|, \\
&\leq \sum_j \omega_j \sqrt{\langle \psi_j | U_2^\dagger U_2 | \psi_j \rangle \langle \psi_j | U_1^\dagger U_1 | \psi_j \rangle}, \\
&= 1.
\end{aligned} \tag{29}$$

Consequently, $\text{Re}[W] \leq 0$, which implies $\text{Im}[I] \geq 0$.

For thermal state, $\rho_0 = e^{-\beta H_0}$, there is another symmetry we should impose, i.e. dynamical Kubo-Martin-Schwinger (KMS) symmetry.

First, KMS condition (Correlation level):

$$\langle A(t_1) B(t_2) \rangle_{\rho_0} = \langle B(t_2) A(t_1 + i\beta) \rangle_{\rho_0}. \tag{30}$$

Proof:

$$\begin{aligned}
\langle A(t_1) B(t_2) \rangle_{\rho_0} &= \text{Tr}(\rho_0 A(t_1) B(t_2)), \\
&= \text{Tr}(e^{-\beta H_0} A(t_1) B(t_2)), \\
&= \text{Tr}(e^{-\beta H_0} A(t_1) e^{\beta H_0} e^{-\beta H_0} B(t_2)), \\
&= \text{Tr}(A(t_1 + i\beta) e^{-\beta H_0} B(t_2)), \\
&= \text{Tr}(e^{-\beta H_0} B(t_2) A(t_1 + i\beta)), \\
&= \langle B(t_2) A(t_1 + i\beta) \rangle_{\rho_0}.
\end{aligned} \tag{31}$$

Then, for the generating functional (omitted spatial indices):

$$\begin{aligned}
e^{W[J_1(t), J_2(t)]} &= \text{Tr}(\rho_0 U^\dagger(+\infty, -\infty; O_2) U(+\infty, -\infty; O_1)), \\
&= \text{Tr} \left(\underbrace{\rho_0 \bar{\mathcal{T}} \left(e^{-i \int_{-\infty}^{+\infty} dt O_2(t) J_2(t)} \right)}_A \underbrace{\mathcal{T} \left(e^{i \int_{-\infty}^{+\infty} dt O_1(t) J_1(t)} \right)}_B \right), \\
&= \text{Tr} \left(\rho_0 \mathcal{T} \left(e^{i \int_{-\infty}^{+\infty} dt O_1(t) J_1(t)} \right) \bar{\mathcal{T}} \left(e^{-i \int_{-\infty}^{+\infty} dt O_2(t+i\beta) J_2(t)} \right) \right), \\
&= \text{Tr} \left(\rho_0 \mathcal{T} \left(e^{i \int_{-\infty}^{+\infty} dt O_1(t) J_1(t)} \right) \bar{\mathcal{T}} \left(e^{-i \int_{-\infty}^{+\infty} dt O_2(t) J_2(t-i\beta)} \right) \right), \\
&\equiv e^{W_T[J_1(t), J_2(t-i\beta)]}.
\end{aligned} \tag{32}$$

We have defined a "reverse" generating functional, which means evolution is reverse along the contour. Suppose the system has several discrete symmetries in microscopic level Θ including time reversal $\mathcal{T}$, sometimes with combinations charge conjugate $\mathcal{C}$ and parity $\mathcal{P}$:

$$W[J_1(t), J_2(t)] = W_T[\Theta J_1(t), \Theta J_2(t)]. \quad (33)$$

Combining (32) and (33), one can get:

$$W_T[\Theta J_1(t), \Theta J_2(t)] = W_T[J_1(t), J_2(t - i\beta)], \quad (34)$$

which means

$$W[J_1(t), J_2(t)] = W[\Theta J_1(t), \Theta J_2(t - i\beta)]. \quad (35)$$

This is a Z_2 operation ($\Theta^2 = \mathbb{I}$). For the action, if we require it has a dynamical KMS symmetry

$$I[\chi_1, J_1; \chi_2, J_2] = I[\tilde{\chi}_1, \tilde{J}_1; \tilde{\chi}_2, \tilde{J}_2], \quad (36)$$

then (35) should be satisfied, where (property 4)

$$\begin{aligned} \tilde{\chi}_1(t, \vec{x}) &= \Theta \chi_1(t, \vec{x}), & \tilde{\chi}_2(t, \vec{x}) &= \Theta \chi_2(t - i\beta, \vec{x}), \\ \tilde{J}_1(t, \vec{x}) &= \Theta J_1(t, \vec{x}), & \tilde{J}_2(t, \vec{x}) &= \Theta J_2(t - i\beta, \vec{x}). \end{aligned} \quad (37)$$

We often use this symmetry in the classical limit ($\hbar \rightarrow 0$ or $\beta\omega \ll 1$):

$$\begin{aligned} \tilde{\chi}_1(t, \vec{x}) &= \Theta \chi_1(t, \vec{x}), & \tilde{\chi}_2(t, \vec{x}) &= \Theta \chi_2(t, \vec{x}) - i\beta \Theta \partial_t \chi_2(t, \vec{x}), \\ \tilde{J}_1(t, \vec{x}) &= \Theta J_1(t, \vec{x}), & \tilde{J}_2(t, \vec{x}) &= \Theta J_2(t, \vec{x}) - i\beta \Theta \partial_t J_2(t, \vec{x}). \end{aligned} \quad (38)$$

We usually redefine the fields: classical χ_r, J_r and noise χ_a, J_a :

$$\begin{aligned} \chi_r &= \frac{\chi_1 + \chi_2}{2}, & \chi_a &= \chi_1 - \chi_2, \\ J_r &= \frac{J_1 + J_2}{2}, & J_a &= J_1 - J_2. \end{aligned} \quad (39)$$

Then the four properties are

1. $I^*[\chi_r, J_r; \chi_a, J_a] = -I[\chi_r, J_r; -\chi_a, -J_a]$.
2. $I[\chi_a = 0] = 0$.
3. $\text{Im}[I] \geq 0$.
4. $I[\chi_r, J_r; \chi_a, J_a] = I[\tilde{\chi}_r, \tilde{J}_r; \tilde{\chi}_a, \tilde{J}_a]$ with

$$\begin{aligned} \tilde{\chi}_r(t, \vec{x}) &= \Theta \chi_r(t, \vec{x}), & \tilde{\chi}_a(t, \vec{x}) &= \Theta \chi_a(t, \vec{x}) - i\beta \Theta \partial_t \chi_r(t, \vec{x}), \\ \tilde{J}_r(t, \vec{x}) &= \Theta J_r(t, \vec{x}), & \tilde{J}_a(t, \vec{x}) &= \Theta J_a(t, \vec{x}) - i\beta \Theta \partial_t J_r(t, \vec{x}), \end{aligned} \quad (40)$$

in the classical limit.

Proof of 4 (take χ for example):

We restore $\hbar$:

$$\begin{aligned} \chi_r &= \frac{\chi_1 + \chi_2}{2}, & \hbar \chi_a &= \chi_1 - \chi_2, \\ \tilde{\chi}_1 &= \Theta \chi_1, & \tilde{\chi}_2 &= \Theta \chi_2 - i\beta \hbar \Theta \partial_t \chi_2. \end{aligned} \quad (41)$$

Then,

$$\begin{aligned}
\tilde{\chi}_r &= \frac{\tilde{\chi}_1 + \tilde{\chi}_2}{2} = \Theta\chi_r - \frac{i\beta\hbar\Theta\partial_t\chi_2}{2}, \\
&= \Theta\chi_r + \mathcal{O}(\hbar), \\
\hbar\tilde{\chi}_a &= \tilde{\chi}_1 - \tilde{\chi}_2 = \Theta\chi_1 - \Theta\chi_2 + i\beta\hbar\Theta\partial_t\chi_2, \\
&= \Theta(\chi_1 - \chi_2) + i\beta\hbar\Theta\partial_t(\chi_r - \frac{\hbar}{2}\chi_a), \\
&= \Theta(\hbar\chi_a) + i\beta\hbar\Theta\partial_t\chi_r - i\beta\hbar\Theta\partial_t(\frac{\hbar}{2}\chi_a), \\
&= \hbar\Theta\chi_a + i\beta\hbar\Theta\partial_t\chi_r + \mathcal{O}(\hbar^2).
\end{aligned} \tag{42}$$

Taking $\hbar \rightarrow 0$, finally we have

$$\tilde{\chi}_r = \Theta\chi_r, \quad \tilde{\chi}_a = \Theta\chi_a + i\beta\Theta\partial_t\chi_r. \tag{43}$$

Appendix A: Derivation of Time Evolution

From (3), one might naively guess that the solution is:

$$U(t, t_0) \stackrel{?}{=} e^{-i\int_{t_0}^t H(\tau)d\tau}, \tag{44}$$

However, since $U(t, t_0)$ must satisfy the group property $U(t, t_0) = U(t, t_1)U(t_1, t_0)$, Eq. (44) is generally incorrect when $[H(t), H(t')] \neq 0$.

We use the BCH formula expansion:

$$e^Xe^Y = e^{X+Y+\frac{1}{2}[X,Y]+\frac{1}{12}[X,[X,Y]]-\frac{1}{12}[Y,[X,Y]]+\dots}. \tag{45}$$

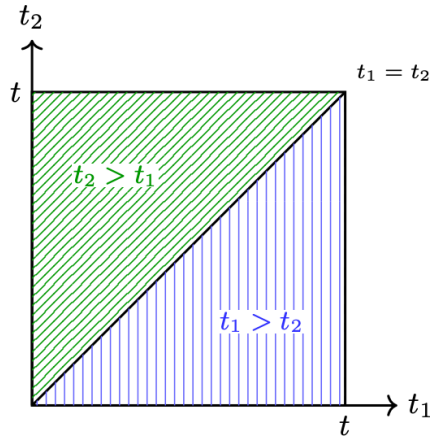
According to (3), $idU(t) = H(t)U(t)dt$. Setting $t_0 \equiv 0$ and $U(t) \equiv U(t, 0)$:

$$\int_0^t dU(t) = -i \int_0^t dt_1 H(t_1)U(t_1). \tag{46}$$

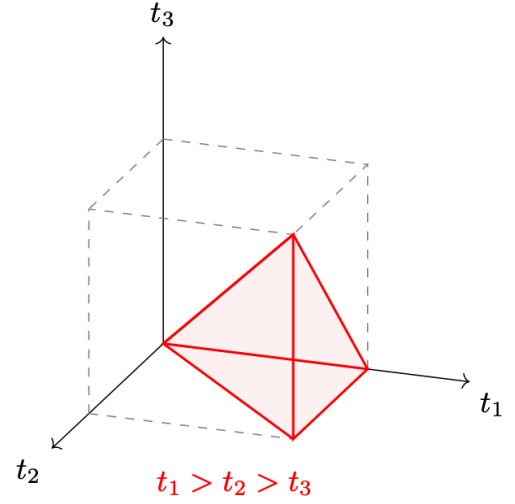
Iterating this integral equation:

$$\begin{aligned}
U(t) &= 1 - i \int_0^t dt_1 H(t_1)U(t_1), \\
&= 1 - i \int_0^t dt_1 H(t_1) \left(1 - i \int_0^{t_1} dt_2 H(t_2)U(t_2) \right), \\
&= 1 - i \int_0^t dt_1 H(t_1) \left(1 - i \int_0^{t_1} dt_2 H(t_2) \left(1 - i \int_0^{t_2} dt_3 H(t_3)U(t_3) \right) \right), \\
&= 1 - iI_1(t) + (-i)^2 I_2(t) + \dots + (-i)^n I_n(t) + \dots.
\end{aligned} \tag{47}$$

We define the n -th order term as $I_n(t) = \int_0^t dt_1 \dots \int_0^{t_{n-1}} dt_n H(t_1) \dots H(t_n)$.



(a) 2D Integration: Symmetry



(b) 3D Case: Simplex Volume

Consider the second-order term:

$$I_2 = \int_0^t dt_1 \int_0^{t_1} dt_2 H(t_1) H(t_2). \quad (48)$$

Geometrically (see diagram), the integration over the time-ordered region $t_1 > t_2$ represents half of the total square area $[0, t] \times [0, t]$. We can rewrite the same integral by relabeling dummy variables ($t_1 \leftrightarrow t_2$) and changing limits to the upper triangular region:

$$I_2 = \int_0^t dt_2 \int_0^{t_2} dt_1 H(t_2) H(t_1). \quad (49)$$

We define the time-ordering operator $\mathcal{T}$ such that operators with later times are placed to the left:

$$\mathcal{T}(H(t_a)H(t_b)) = \begin{cases} H(t_a)H(t_b), & \text{if } t_a > t_b, \\ H(t_b)H(t_a), & \text{if } t_b > t_a. \end{cases} \quad (50)$$

Summing (48) and (49):

$$\begin{aligned} 2I_2 &= \int_0^t dt_1 \int_0^{t_1} dt_2 H(t_1) H(t_2) + \int_0^t dt_2 \int_0^{t_2} dt_1 H(t_2) H(t_1), \\ &= \int_0^t dt_1 \int_0^t dt_2 \mathcal{T}(H(t_1)H(t_2)). \end{aligned} \quad (51)$$

Thus:

$$I_2 = \frac{1}{2!} \int_0^t dt_1 \int_0^t dt_2 \mathcal{T}(H(t_1)H(t_2)). \quad (52)$$

This logic generalizes to n dimensions. The integral over the simplex $t > t_1 > t_2 > \dots > t_n > 0$ is equal to $1/n!$ times the integral over the hypercube:

$$I_n = \frac{1}{n!} \int_0^t dt_1 \dots \int_0^t dt_n \mathcal{T}(H(t_1) \dots H(t_n)). \quad (53)$$

Finally:

$$\begin{aligned}
U(t) &= 1 + \sum_{n=1}^{\infty} \frac{(-i)^n}{n!} \int_0^t dt_1 \cdots \int_0^t dt_n \mathcal{T}(H(t_1) \cdots H(t_n)), \\
&= \mathcal{T} \left(e^{-i \int_0^t H(\tau) d\tau} \right).
\end{aligned} \tag{54}$$

Appendix B: Path Integral for the Propagator

In quantum mechanics, we calculate the propagator $K(x_f, t_f; x_i, t_i)$ from (x_i, t_i) to (x_f, t_f) :

$$K(x_f, t_f; x_i, t_i) = \langle x_f | e^{-\frac{i}{\hbar} H(t_f - t_i)} | x_i \rangle \tag{55}$$

Assume the Hamiltonian is $\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$. We explicitly retain $\hbar$ and the operator 'hat' notation.

We perform time slicing with $\epsilon = (t_f - t_i)/N$ and $t_j = t_i + j\epsilon$ ($j = 0, 1, \dots, N$):

$$e^{-\frac{i}{\hbar} \hat{H}(t_f - t_i)} = \underbrace{e^{-\frac{i}{\hbar} \hat{H}\epsilon} \cdots e^{-\frac{i}{\hbar} \hat{H}\epsilon}}_{N \text{ times}}. \tag{56}$$

We insert the identity operator $\int dx |x\rangle \langle x| = \mathbb{I}$ at every time slice ($N - 1$ times), labeling coordinates $x_1, \dots, x_{N-1}$:

$$\begin{aligned}
K &= \int dx_1 \cdots dx_{N-1} \langle x_f | e^{-\frac{i}{\hbar} \hat{H}\epsilon} | x_{N-1} \rangle \langle x_{N-1} | e^{-\frac{i}{\hbar} \hat{H}\epsilon} | x_{N-2} \rangle \cdots \langle x_1 | e^{-\frac{i}{\hbar} \hat{H}\epsilon} | x_i \rangle, \\
&= \int \left(\prod_{j=1}^{N-1} dx_j \right) \prod_{j=0}^{N-1} \langle x_{j+1} | e^{-\frac{i}{\hbar} \hat{H}\epsilon} | x_j \rangle.
\end{aligned} \tag{57}$$

Using the Trotter decomposition $e^{\epsilon(A+B)} \approx e^{\epsilon A} e^{\epsilon B}$ (valid to order $\mathcal{O}(\epsilon^2)$), we separate the kinetic and potential terms:

$$\begin{aligned}
\langle x_{j+1} | e^{-\frac{i}{\hbar} (\hat{T} + \hat{V})\epsilon} | x_j \rangle &\approx \langle x_{j+1} | e^{-\frac{i}{\hbar} \hat{V}\epsilon} e^{-\frac{i}{\hbar} \hat{T}\epsilon} | x_j \rangle, \\
&= e^{-\frac{i}{\hbar} V(x_{j+1})\epsilon} \langle x_{j+1} | e^{-\frac{i}{\hbar} \frac{\hat{p}^2}{2m}\epsilon} | x_j \rangle.
\end{aligned} \tag{58}$$

To calculate the kinetic term, we insert the momentum basis $\int dp |p\rangle \langle p| = \mathbb{I}$:

$$\begin{aligned}
\langle x_{j+1} | e^{-\frac{i}{\hbar} \frac{\hat{p}^2}{2m}\epsilon} | x_j \rangle &= \int dp_j \langle x_{j+1} | e^{-\frac{i}{\hbar} \frac{\hat{p}^2}{2m}\epsilon} | p_j \rangle \langle p_j | x_j \rangle, \\
&= \int dp_j e^{-\frac{i}{\hbar} \frac{p_j^2}{2m}\epsilon} \langle x_{j+1} | p_j \rangle \langle p_j | x_j \rangle, \\
&= \int dp_j e^{-\frac{i}{\hbar} \frac{p_j^2}{2m}\epsilon} \left(\frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i}{\hbar} p_j x_{j+1}} \right) \left(\frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar} p_j x_j} \right), \\
&= \frac{1}{2\pi\hbar} \int dp_j \exp \left\{ \frac{i}{\hbar} \left[p_j (x_{j+1} - x_j) - \frac{\epsilon}{2m} p_j^2 \right] \right\}.
\end{aligned} \tag{59}$$

Here we used the plane wave formula $\langle x | p \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i}{\hbar} p x}$. Using the Gaussian integral formula:

$$\int_{-\infty}^{\infty} e^{-ax^2 + bx} dx = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}}, \tag{60}$$

with $a = \frac{i\epsilon}{2m\hbar}$ and $b = \frac{i(x_{j+1} - x_j)}{\hbar}$, we get:

$$\langle x_{j+1} | e^{-\frac{i}{\hbar} \hat{T} \epsilon} | x_j \rangle = \underbrace{\sqrt{\frac{m}{2\pi i \hbar \epsilon}}}_A \exp \left[\frac{i}{\hbar} \frac{m}{2} \frac{(x_{j+1} - x_j)^2}{\epsilon} \right]. \quad (61)$$

Here A is the measure factor. The propagator for a short time step is:

$$\langle x_{j+1} | e^{-\frac{i}{\hbar} \hat{H} \epsilon} | x_j \rangle \approx A \cdot \exp \left[\frac{i}{\hbar} \epsilon \left(\frac{m}{2} \left(\frac{x_{j+1} - x_j}{\epsilon} \right)^2 - V(x_{j+1}) \right) \right]. \quad (62)$$

We observe that:

- $\frac{x_{j+1} - x_j}{\epsilon} \approx \dot{x}$ (Velocity).
- $\frac{1}{2} m \dot{x}^2 - V(x) = L(x, \dot{x})$ (Lagrangian).

Thus:

$$K = \lim_{N \rightarrow \infty} \int dx_1 \dots dx_{N-1} A^N \exp \left[\frac{i}{\hbar} \sum_{j=0}^{N-1} \epsilon \left(\frac{m}{2} \left(\frac{x_{j+1} - x_j}{\epsilon} \right)^2 - V(x_j) \right) \right]. \quad (63)$$

Taking the continuum limit ($N \rightarrow \infty, \epsilon \rightarrow 0$):

$$\sum_{j=0}^{N-1} \epsilon L(x_j, \dot{x}_j) \longrightarrow \int_{t_i}^{t_f} dt L(x(t), \dot{x}(t)) = S[x(t)]. \quad (64)$$

We replace the multiple integral with the path integral measure:

$$\mathcal{D}x(t) \equiv \lim_{N \rightarrow \infty} A^N \prod_{j=1}^{N-1} dx_j. \quad (65)$$

The final result is:

$$K(x_f, t_f; x_i, t_i) = \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x(t) e^{\frac{i}{\hbar} S[x(t)]}, \quad (66)$$

where

$$S[x(t)] = \int_{t_i}^{t_f} \left[\frac{1}{2} m \dot{x}^2 - V(x) \right] dt. \quad (67)$$

Finally, we introduce the source $J(x)$ and generalize to quantum field theory:

$$Z[J] = \int \mathcal{D}\phi \exp \left\{ i \int d^4x (\mathcal{L}[\phi] + J(x)\phi(x)) \right\}. \quad (68)$$